

Assignment 1

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1 Task 1

For task 1 we chose to use a Sigma of 2 and a threshold value of 128.

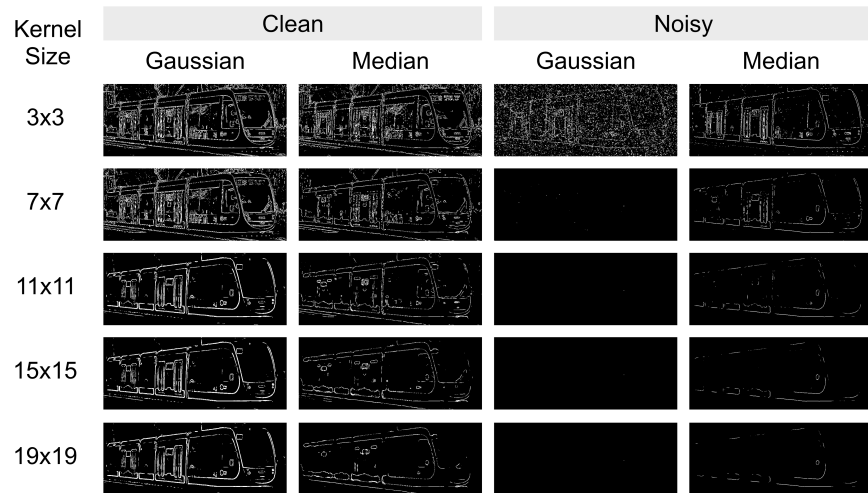


Figure 1: Images generated for task 1, using a threshold of 128 and a Sigma of 2 for the Gaussian filter.

1.1 Clean Images

The Gaussian filter's kernel size initially has a large effect on the edge detection, but it quickly becomes imperceptible, with the later results being identical. This is because increasing the kernel size does not affect the sigma value, and the distribution tails remain small. The median filter on the other hand shows a

more gradual and perceptible decrease in detected edges, although the decrease remains front-loaded. This is due to the median filter's strength being directly caused by its size, with a larger kernel averaging over a larger area.

1.2 Noisy Images

The relationship between kernel size and effect remains present in the noisy images, with the Gaussian filter having a very strong ramp, while the median filter is more gradual. More interestingly is the difference in the way the two filters interact with the noise; whereas the median filter can effectively diminish noise, the edge detection remains noisy despite the Gaussian filter at a small kernel size, and fails entirely at higher kernel sizes. This comes down to two factors: (1) the noise causes ImageA-noisy to already have maximum contrast such that it was not modified by the contrast adjustment, and (2) the gaussian filter is vulnerable to outliers as it uses a weighted average. This means the filtered image before edge detection is dominated by noise, and the actual information is drowned out. This could be avoided by using modified contrast adjustment and filtering out the spikes at the contrast extremes.

2 Task 2: Grayscale Erosion Analysis

As seen in the images (See figure 2), the size of lighter areas is reduced when a larger structuring element is used. As we can see, a larger structuring element causes the image to be less bright by introducing more dark spots and reducing bright spots.

To support this claim the number of unique pixel intensity values as well as average pixel intensity values have been outlined in a graphs (see graph 3).

This is to be expected, as grayscale erosion makes each pixel the minimum value within the bounds of the structuring element used. The outcome is a less bright image with more dark spots.

The structuring elements used for the experiments consist of all 0 values, with the matrix sizes being 3×3 , 7×7 , 11×11 , 15×15 and 19×19 .



(a) Original picture



(b) 3x3 grayscale erosion



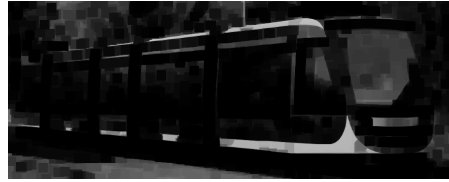
(c) 7x7 grayscale erosion



(d) 11x11 grayscale erosion

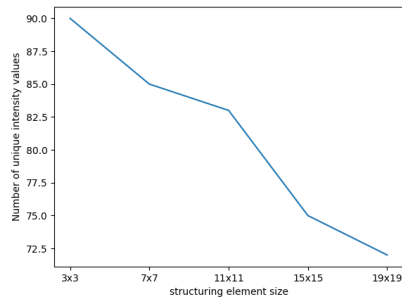


(e) 15x15 grayscale erosion

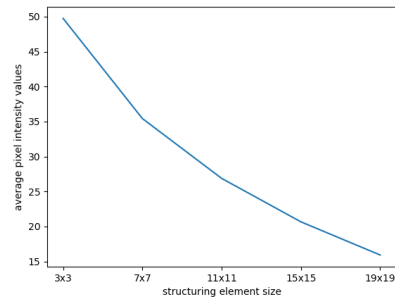


(f) 19x19 grayscale erosion

Figure 2: The effect of grayscale erosion with different sizes of structuring elements generated in task 2



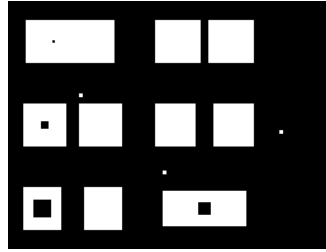
(a) unique pixel intensity values



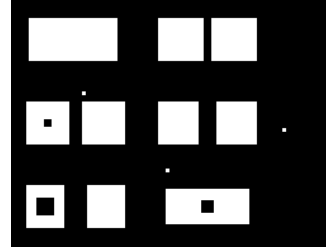
(b) average pixel intensity values

Figure 3: Graphs about pixel intensity values generated in task 2

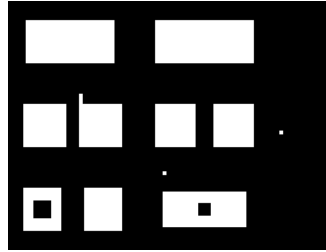
3 Task 3: Binary Closing analysis



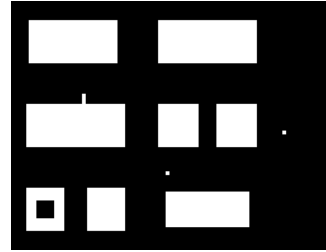
(a) Original image



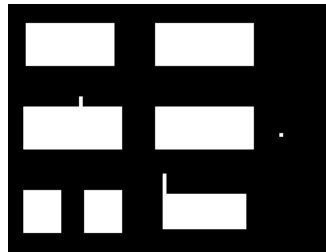
(b) Binary Closing using a 3x3 structuring element



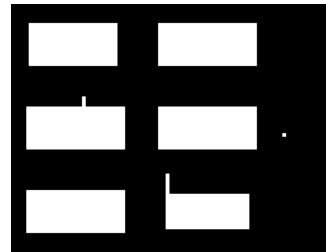
(c) Binary Closing using a 7x7 structuring element



(d) Binary Closing using a 11x11 structuring element



(e) Binary Closing using a 15x15 structuring element



(f) Binary Closing using a 19x19 structuring element

Figure 4: The effect of binary closing with different sizes of structuring elements generated in task 3

As we can see from the images (see figure 4), An increase in the size of the structuring element causes bigger holes in the white blocks to get filled in.

In figure 4b we can see that the small hole in the upper left white square is filled, in figure 4c we can see the hole in the left middle square gets filled, as well as the square right of it now being connected to the small white dot above it. This trend continues onto bigger holes as the structuring element grows in size.

This occurs because a structuring element can fill holes that are small enough

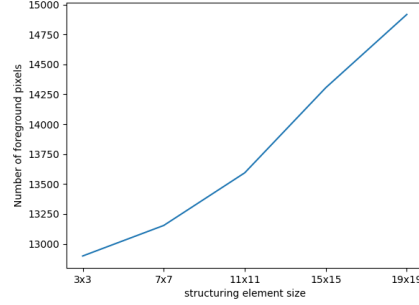


Figure 5: Comparison of the size of structuring element used compared to the number of foreground pictures.

relative to its size, meaning that a larger structuring element is capable of filling in bigger holes.

The derived graph (see figure 5) also shows this, signaling that an increase in structuring element size corresponds to an increase in white pixels for this specific picture.

4 Choice Tasks

4.1 Edge Sharpening

We decided to implement unsharp masking edge sharpening as we already had most of the required functions. We generate a blurred image using the existing controls for Task 1, then subtract it from the original image to find the difference between these and obtain a mask, adding that back to the original image. We also only check if the absolute difference between the sharp and blurred image is greater than a threshold, and multiply the mask by a strength factor, both of which can be set by the GUI.



(a) Unsharpened Image A

(b) Sharpened Image A

Figure 6: Original Image A sharpened with Gaussian filter of Kernel size 21 and Sigma of 10, a threshold of 10, and a sharpening factor of 1 (100%)

4.2 Histogram Checking

We implemented a function to generate a histogram with the coordinates of all pixels of a particular intensity value. We then find the minimum and maximum intensity. If they are not 0 and 255 respectively we know the image does not have the maximum possible contrast. We then check if there are any values missing between maximum and minimum, and if there are, we know the dynamic range is also not at the maximum value.

4.3 Histogram Matching

Using the histogram generation function we defined in the previous choice task, we implemented a histogram matching function. A reference image can be loaded in. The function will step through each intensity value of the input and also the output, creating a reference intensity for each input value.



(a) Original Image A



(b) Matched to histogram of Task 3 image